Analytical Cumulative Examination
23rd April 1993

One of the most widespread methods of instrumental analysis is INAA, instrumental neutron activation analysis.

1. Describe the basic principles of the method (20).

2. Show that the activity, A, of a nuclide produced in determining an element whose atomic abundance is N_t after a period t in the reactor is given by:

$$A = \sigma N_t \phi [1 - \exp(-\lambda t)]$$

where σ is the reaction cross section for thermal neutron capture, λ is the decay constant and ϕ is the neutron flux (20)

3. Sketch the curve of A against t for (10):

(1) a nuclide in which $t \gg 1/\lambda$
(2) a nuclide in which $t \ll 1/\lambda$

4. Calculate the activity of ^{27}Mg and ^{24}Na (20):

(1) at the end of irradiation
(2) 24 hours after the irradiation

5. Discuss the implications of your results for the analysis of Na and Mg by this method (15).

6. Na and Mg are normally present in rock as silicates, but Si and O cannot be determined by this method. Explain why and suggest alternative ways whereby they might be determined (15).

An excerpt from the Chart of the Nuclides is attached.